

Low-Dimensional Generative Models of Human Interception Movements

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Abstract

We investigate whether a compact representation inferred from a participant's interception trials can generate the distribution of that participant's remaining movements. We evaluate a conditional variational autoencoder (CVAE) and four comparator families on 4,732 trials from 28 participants, using four participant-held-out folds and three neural training seeds. The input preserves the interval from target motion to the end of the recorded reach; physical timing is predicted separately from the phase-normalized trajectory. Spline+PCA provides lower reconstruction error at matched latent dimension, whereas CVAE provides lower timing error and several lower generated-distribution distances. A common-head sensitivity analysis reduces, but does not eliminate, the CVAE's initiation-time advantage. Participant-level behavioural probes recover some between-participant differences but fail on many targets. Explicit task conditioning and minimum-jerk component fidelity remain limited. The findings support task-dependent benefits of a learned representation, not a uniformly superior model or a validated two- or three-parameter account of individual strategy.

1 Introduction

Interception requires predicting a moving target while allowing online corrections [1]. Compact models of individual movement variation could support future motor-assessment research, but clinical inference is outside this study. Following work on individual motor signatures [2], our question is whether a low-dimensional representation of recorded trajectories can support participant-specific generation, rather than merely identify participants. The requested compact interface is two or three latent coordinates; $n=8$ serves as a higher-capacity comparison.

We distinguish three tasks: reconstruction of an observed trajectory; prediction of its physical timing from shape; and generation of new trials after enrollment from a subset of a held-out participant's recordings. A successful reconstruction does not establish the third task. Our machine-learning objective is therefore evaluated against both classical compression and non-personalized stochastic baselines.

2 Data and temporal representation

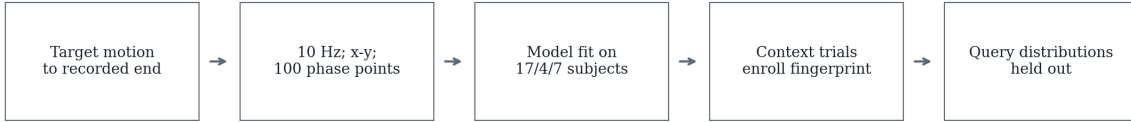
The experiment records three-dimensional finger position at 240 Hz using an electromagnetic tracker while a participant reaches toward an interception region on a tabletop. Target trajectories run at 60 Hz. We analyze condition 2 (free eye movements). Of 4,763 available condition-2 recordings, the canonical rules remove four timeouts and 27 excessively late arrivals, retaining 4,732 trials from 28 participants. The source project description lists 29 participants; the analyzed cohort is the 28 with available condition-2 recordings. Trial filenames encode condition, start-position/speed category, side and repetition. MAT metadata supplies exact target speed and event times. Repetition and outcome labels are not encoder inputs.

Marker 5 identifies target *appearance*, not motion. Time zero is the MAT-derived target-motion event, which follows appearance by 0.20-0.50 s (median 0.35 s). Tracker timing uses frame-counter intervals at 240 Hz. The retained window ends at the last tracker sample, an arrival proxy rather than the MAT `pressedTime` itself. This proxy precedes the MAT event by 26.2 ms on average (SD 3.6 ms). Figure A3 documents the event audit; synchronization requires confirmation from Prof. Friedman.

Position is fourth-order, zero-phase Butterworth filtered at 10 Hz, translated to the window origin and resampled to 100 phase points. Lateral x and forward y are modeled; z is omitted from the trajectory output. The canonical movement-onset label is the first post-go 3-D speed crossing above 5 tracker units/s sustained for three frames. Initiation time is go-to-onset; movement time is onset-to-recording-end. Both physical

times are withheld from the encoder. The primary go-to-end input retains post-go waiting; an earlier movement-only representation was explored but is not part of the present model comparison.

The late-arrival cutoff is 1 s beyond the target's interception window. A 'Too early' outcome is not treated as premature departure: recorded early arrivals after the go event remain included. The retained cohort includes 326 'Too late' and 1,352 'Not fixating on the dot enough!!!' outcomes. Other quality flags are audits, not additional exclusions. Tracker coordinates are reported in tracker units because their physical calibration is unconfirmed.



Four outer participant folds; each participant tested once; neural seeds 42/43/44

Figure 1. Participant-held-out protocol. Context and query trials are disjoint within each test participant; no query trajectory determines the enrollment fingerprint.

3 Models and enrollment

The CVAE [3,4] maps a 100 x 2 trajectory and a five-element condition vector (three start-category indicators, side and exact target speed) through two 256-unit ReLU layers to a diagonal-Gaussian posterior. A corresponding two-layer decoder takes a latent draw and the condition vector, and outputs position and two timing channels. Training uses participant-balanced sampling, Adam (learning rate 0.001; batch size 64), gradient clipping at 5, validation early stopping (patience 25) and a 200-epoch limit. All normalization statistics are fitted to training data.

The objective is a summed standardized position squared error plus 20 times the summed standardized log-timing squared error and a KL penalty to $N(0, I)$. The KL coefficient increases linearly from 0 to 1 over 50 epochs. Timing is transformed as $\log(t+0.001)$ and inverted by $\max(\exp(u)-0.001, 0)$. These are weighted reconstruction objectives, not calibrated likelihoods for the reported behavioural distributions. KL regularization does not guarantee semantic independence, identifiable latent axes or meaningful averages for every participant.

Table 1. Model families. Equal latent dimension does not equalize parameter count, objective or timing-head capacity.

Model	Representation and comparison
CVAE; n=2,3,4,8	Variational trajectory code with task condition; joint shape and timing objective.
Spline+PCA; n=2,3,4,8	Cubic B-spline fit (five knots), training-fitted PCA and validation-tuned Ridge log-timing head; classical compression comparator.
Conditional AE; n=3,8	Same conditional architecture without stochastic bottleneck or KL term.
Unconditional VAE; n=3,8	Same variational architecture with task condition suppressed.
Condition Ridge	Task metadata alone predicts shape and timing; training-stratum residual resampling supplies stochastic variation without a personal fingerprint.

For a held-out participant s , the fingerprint is the average posterior mean over context trials: $\theta_s = \text{mean}\{\mu(x, c) : (x, c) \text{ in context}_s\}$. Generation samples z from $N(\theta_s, \Sigma_{\text{train}})$, using a shared covariance estimated from within-participant training variability and posterior uncertainty. The decoder generates 120 trials per participant. Query task conditions are sampled to match the evaluated condition mixture; query trajectories and timings are not used for enrollment. Thus generation is conditional on a known task mixture, not zero-shot prediction of a new participant. The spline generator similarly uses context-mean coefficients with training-derived variation. A shared latent covariance does not imply equal output variance under a nonlinear decoder.

4 Evaluation

Four fixed outer folds each contain 17 training, four validation and seven test participants. Every participant is tested once. Context/query allocation is approximately 50/50, stratified by task where possible. Neural seeds 42, 43 and 44 vary training randomness; the fold allocation is fixed. The matrix contains 48 CVAE, 24 conditional-AE, 24 unconditional-VAE, 16 spline and 12 condition-Ridge evaluations. This covers all specified folds and seeds, not all possible participant splits. Training sets overlap across folds; there is no independent acquisition cohort.

For participant s , trajectory MSE averages squared raw-coordinate reconstruction error over that participant's trials, phase points and two axes. Timing MAE averages absolute error over trials, then participants receive equal weight. Reconstruction uses posterior means, not sampled posterior noise. $R^2 = 1 - \text{sum}(y - \hat{y})^2 / \text{sum}(y - \text{mean}(y))^2$. Trial-pooled timing R^2 , R^2 of participant means, and participant-summary probe R^2 have different observational units and are reported separately in the accompanying tables.

Distribution fidelity uses two-sample KS statistics on 11 features: initiation and movement time, peak speed, relative time to peak speed, path length, straight-line distance, curvature, lateral deviation, speed-peak count and x/y endpoints. The reported mean KS weights features and participants equally; it is an effect-size summary, not a combined hypothesis test. Speed peaks use a 10% prominence and 50 ms separation heuristic and are distinct from fitted minimum-jerk components. Movement-derived features are computed on the extracted movement interval using the same feature routine for recorded and generated paths. Generated intervals are recovered from predicted initiation and movement time.

Multivariate distances use one training-only feature reference per fold shared by every model. Continuous coordinates are scaled by training SD; count scales are at least one count and constant features use scale one. RBF bandwidth is fixed from up to 512 training examples. We report the unbiased squared MMD estimator (which can be slightly negative at finite sample size) and energy $E = 2 \text{ mean}\|X - Y\| - \text{mean}\|X - X'\| - \text{mean}\|Y - Y'\|$, including within-sample diagonal terms. Lower discrepancy is preferable. KS testing for discrete counts is not used as a continuous-null proof of agreement.

Paired comparisons average seeds within each participant before two-sided Wilcoxon tests. Benjamini-Hochberg correction covers the 56 model-endpoint comparisons as one family; the eight added timing-head comparisons form a separate sensitivity family. Cross-fold participant dependence and the small cohort limit population-level inference. Closed-set enrollment assigns each query code to the closest of seven context centroids; nominal chance is 1/7. K-means is an in-sample descriptive analysis, not held-out classification.

5 Results

5.1 Reconstruction and generated distributions

Table 2. Out-of-fold participant-balanced means; neural results average three seeds. MSE is in tracker units squared. Timing uses the original heads; Table 4 examines head capacity and calibration. Enrollment is a closed-set task, not zero-shot identity prediction.

Model	n	MSE	Init. MAE (ms)	Move MAE (ms)	Mean KS	MMD ²	Enroll. (%)
Condition Ridge	-	1.234	98.5	72.7	0.277	0.126	-
Conditional AE	3	0.170	37.8	44.1	0.288	0.141	39.0
Conditional AE	8	0.026	39.9	42.7	0.259	0.111	46.5
CVAE	3	0.177	37.8	43.4	0.278	0.131	40.9
CVAE	8	0.032	34.8	44.2	0.219	0.067	50.7
Spline+PCA	3	0.097	56.2	51.2	0.325	0.181	36.4
Spline+PCA	8	0.023	55.6	50.2	0.255	0.115	48.6
Unconditional VAE	3	0.148	37.9	42.9	0.285	0.130	40.0
Unconditional VAE	8	0.033	36.0	42.2	0.215	0.064	50.8

Spline+PCA has lower observed-trajectory MSE than CVAE at $n=3$ (0.097 versus 0.177) and $n=8$ (0.023 versus 0.032). CVAE has lower original-head timing MAE, mean KS and multivariate discrepancy at those dimensions (Table 3). These comparisons establish an endpoint-dependent tradeoff; they do not show that VAE is intrinsically superior to spline approximation. Representative reconstructions and generated paths appear in Figure A1.

Figure 2. Out-of-fold benchmark endpoints (28 held-out participants)

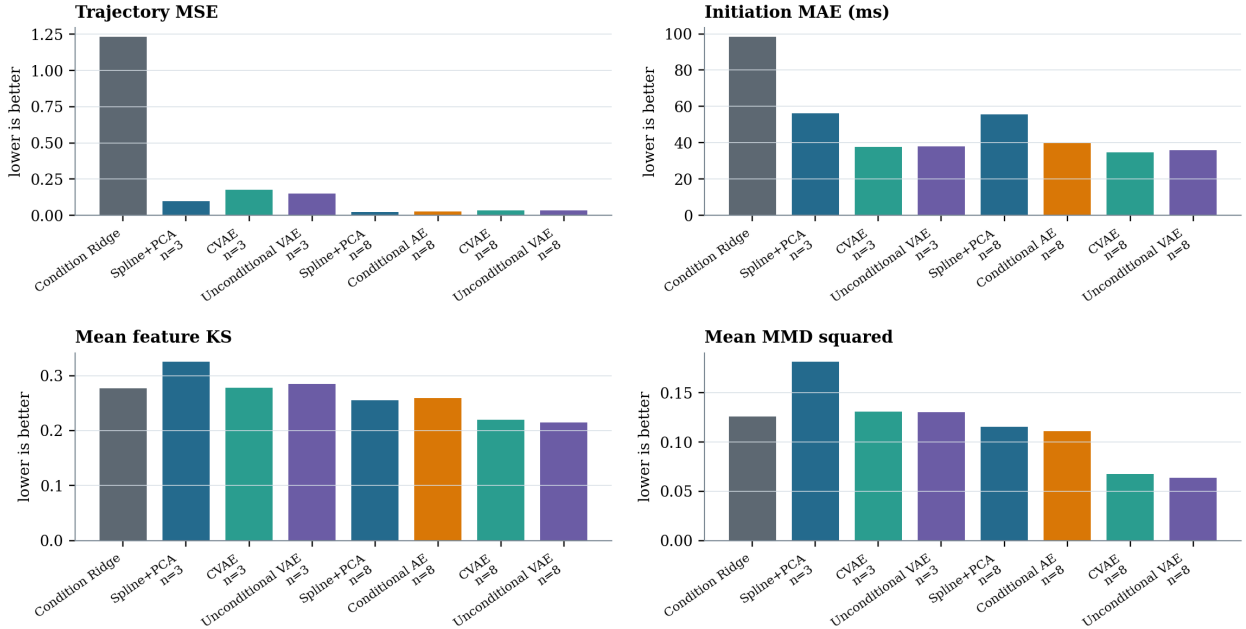


Figure 2. Selected benchmark endpoints. All panels are errors or discrepancies; smaller values indicate better performance for that particular task.

Table 3. Paired CVAE-spline contrasts, adjusted within the complete 56-comparison family. The lower MSE belongs to spline; timing is in ms.

n	Endpoint	CVAE	Spline	CVAE lower /28	Adjusted p
3	Trajectory MSE	0.177	0.097	6	<0.0001
3	Initiation MAE	37.8	56.2	27	<0.0001
3	Movement MAE	43.4	51.2	24	<0.0001
3	Mean KS	0.278	0.325	26	<0.0001
3	MMD squared	0.131	0.181	24	<0.0001
8	Trajectory MSE	0.032	0.023	4	0.0008
8	Initiation MAE	34.8	55.6	27	<0.0001
8	Movement MAE	44.2	50.2	23	0.0013
8	Mean KS	0.219	0.255	23	0.0001
8	MMD squared	0.067	0.115	25	<0.0001

At $n=3$, CVAE does not improve mean KS, MMD or energy over the condition-only baseline after correction. At $n=8$, all three distribution distances are lower than that baseline. Relative to unconditional VAE, the results are mixed: at $n=3$ unconditional VAE has lower MSE while CVAE has lower mean KS; at $n=8$ unconditional VAE has lower energy distance. Most other endpoints do not distinguish these two families. Hence explicit conditioning is not uniformly beneficial, and a useful personal representation cannot be inferred from reconstruction alone.

Figure 3. CVAE capacity and fingerprint enrollment

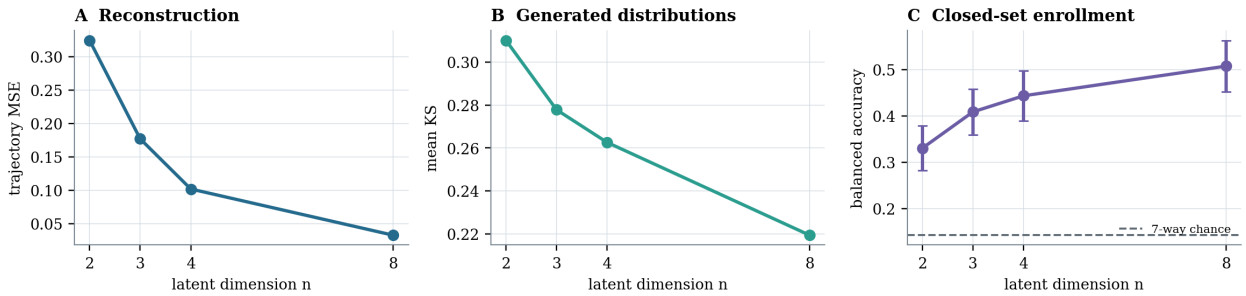


Figure 3. CVAE capacity and closed-set enrollment. Accuracy error bars describe fold/seed spread, not a population confidence interval. Increasing n improves mean reconstruction and distribution fidelity, but does not prove interpretable coordinate semantics.

5.2 Timing-head sensitivity

The original CVAE timing head is nonlinear, whereas spline uses linear Ridge. Direct exponentiation of a squared-log-error prediction also need not optimize physical-time MAE. We therefore add two post-review sensitivities, without selecting on test errors: (i) one positive multiplicative factor per timing endpoint fitted to validation MAE; and (ii) the same two-layer 64-unit MLP timing head trained on frozen CVAE or spline codes plus conditions. The latter uses train-fitted input/log-target scaling, participant weights and validation early stopping (up to 400 epochs, patience 40). It comprises 48 additional small head fits, not retraining of the generative models.

Table 4. Timing sensitivities. Paired participants are seed-averaged; correction covers these eight tests. Calibration changes only timing predictions for this sensitivity, not the main generator.

Head comparison	n	Endpoint	CVAE ms	Spline ms	Adjusted p
Common MLP	3	initiation	41.4	45.8	0.0076
Common MLP	3	movement	43.9	46.1	0.1872
Common MLP	8	initiation	42.7	47.9	0.0423
Common MLP	8	movement	44.8	44.8	0.8314
Validation calibration	3	initiation	37.2	44.4	<0.0001
Validation calibration	3	movement	44.3	50.9	<0.0001
Validation calibration	8	initiation	34.5	53.6	<0.0001
Validation calibration	8	movement	44.7	50.5	0.0001

With the common MLP, initiation MAE is 41.4 versus 45.8 ms at $n=3$ and 42.7 versus 47.9 ms at $n=8$, favouring CVAE; movement-time differences are not significant. The smaller gaps show that the original timing comparison was partly sensitive to head choice. CVAE codes were also learned with timing supervision whereas spline codes were not, so this does not isolate representation geometry alone. Validation calibration is an MAE sensitivity, not a test-set oracle, a smearing correction or evidence assigning a percentage of the advantage to a specific cause.

5.3 Participant-level behavioural prediction

A separate validation-tuned Ridge probe predicts 14 query-summary targets (mean and SD of seven behaviours) from context fingerprints. It is trained on training participants, selects regularization on validation participants and refits on both before testing. The comparison constant is the original 17-training-participant mean. Table 5 includes negative results. Pooled out-of-fold R^2 uses all 28 held-out summary predictions per seed; mean fold R^2 averages seven-participant scores and is not the same statistic.

Table 5. CVAE mean-fingerprint probes, averaging each stated R-squared statistic across available seeds. Speed-peak count is the heuristic feature, not minimum-jerk order.

Query-summary target	n=3 pooled R^2	n=3 fold R^2	n=8 pooled R^2	n=8 fold R^2
Curvature index (mean)	-0.242	-5.660	-0.215	-5.672
Curvature index (sd)	-0.123	-1.204	-0.141	-1.499
Initiation time (mean)	0.723	0.661	0.707	0.655
Initiation time (sd)	-0.011	-0.188	-0.032	-0.803
Lateral deviation (mean)	0.060	-0.215	0.319	0.273
Lateral deviation (sd)	-0.426	-2.842	-0.214	-1.476
Movement time (mean)	0.026	-2.934	0.113	-2.337
Movement time (sd)	-0.109	-0.357	-0.105	-0.599
Speed-peak count (mean)	-0.305	-4.611	-0.267	-5.111
Speed-peak count (sd)	-0.354	-1.070	-0.257	-0.964
Path length (mean)	-0.388	-2.573	-0.130	-2.050
Path length (sd)	-0.099	-1.376	-0.114	-1.647
Peak speed (mean)	0.356	0.346	0.320	0.303
Peak speed (sd)	-0.103	-0.268	0.007	-0.135

For $n=3$, pooled R^2 is 0.723 for mean initiation time and 0.356 for mean peak speed. Their MAEs are 14.8 ms versus a 32.2 ms training-mean baseline, and 7.74 versus 12.08 tracker units/s, respectively. Ten of 14

pooled R^2 values are negative; 12 of 14 mean fold scores are negative. Thus broad recovery of a participant's behavioural distribution parameters is not established. All model-family probes and available mean-plus-SD ablations are included in the downloadable tables; adding SD increases fingerprint dimension and is not equivalent to a two- or three-number interface.

5.4 Conditioning, clustering and latent associations

The in-sample trajectory K-means analysis gives $ARI=0.052$, with permutation $p=0.00498$ (200 permutations; minimum resolution $1/201$). This indicates weak non-random structure, not clean subject clusters. Context enrollment is a distinct held-out task (Table 2). Neither result guarantees that latent means preserve all participant-specific variability.

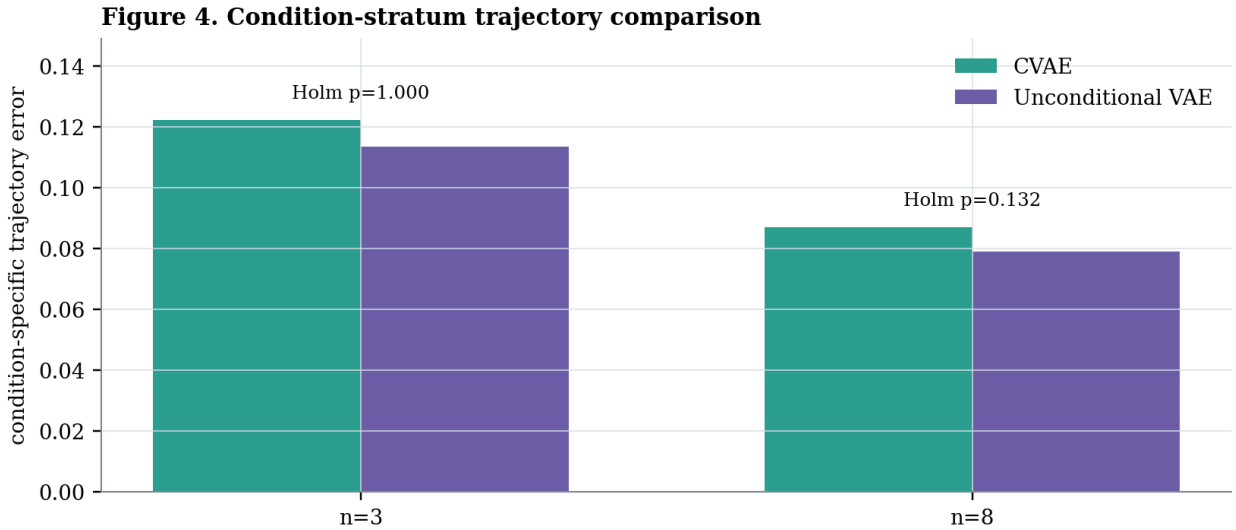


Figure 4. Matched condition-stratum trajectory diagnostic with fixed fingerprints and latent draws. Holm-adjusted $p=1.000$ ($n=3$) and 0.132 ($n=8$); none of 44 feature-level comparisons survives BH correction. This diagnostic does not prove that the decoder ignores its condition input.

The dashboard's latent-feature heatmap is a within-run Spearman association under a specified sampling distribution and fixed task condition. Such correlations are descriptive, can reflect correlated latent draws, and do not establish causal control or coordinate identity across seeds. Figure A4 provides a reproducible example. We do not equate an axis with a cognitive strategy.

6 Minimum-jerk analysis

As a secondary kinematic analysis, recorded and generated movement velocities are fitted with one to four overlapping planar minimum-jerk components [5]. Each component has onset, duration and x/y displacement. This is an adaptation of Prof. Friedman's repository [6], not an unchanged execution: optimization, bounds and the tangential-speed objective differ. The primary model-order rule selects the smallest order with normalized error at most 0.05, then at most 0.10, otherwise the minimum-error candidate. BIC is an alternative. Component number is not a validated classification into wait-then-go versus corrective strategy.

Figure 5. Minimum-jerk component analysis is model-order sensitive

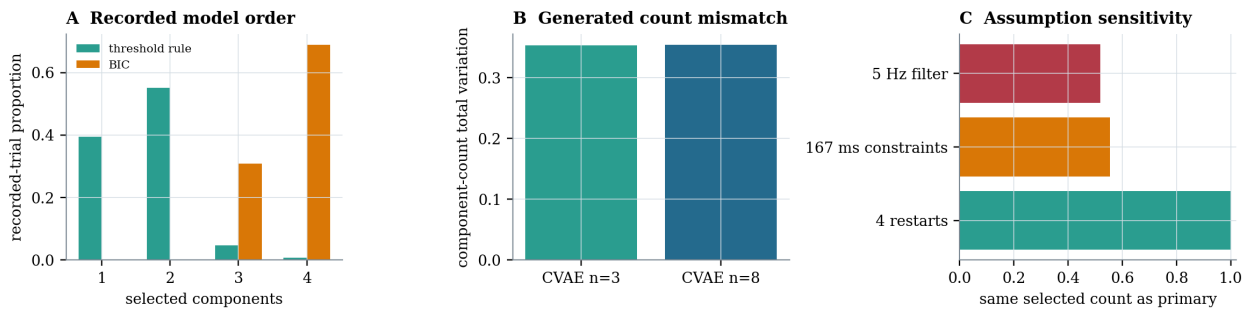


Figure 5. Component counts, generated count discrepancy and sensitivity to fitting assumptions. The generated analysis uses only 10 samples per participant per seed at $n=3$ and $n=8$ (1,680 generated fits overall).

Generated count total variation is 0.353 ($n=3$) and 0.355 ($n=8$); none of nine paired dimension comparisons survives BH correction. On 56 sampled recorded trials, additional optimizer restarts preserve threshold-selected counts in 100%, but the 167 ms constraint and 5 Hz filter preserve them in only 55.4% and 51.8%. Counts therefore depend materially on analysis settings.

Table 6. Empirical finite-sample reference at the actual generated sample size. Each of 500 replicates draws two independent samples from each participant's empirical query distribution, matches sample sizes, and averages three seed-like draws before pooling participants. This is a plug-in diagnostic, not a formal null p-value or hard KS floor.

Discrepancy	Observed $n=3$	Observed $n=8$	Sampling reference mean
Fit error KS	0.318	0.314	0.267
First duration KS	0.450	0.401	0.267
First amplitude KS	0.433	0.411	0.267
Secondary amplitude fraction KS	0.422	0.394	0.242
Overlap KS	0.405	0.392	0.241
Count total variation	0.353	0.355	0.151
Count JSD	0.156	0.166	0.044

The empirical reference accounts for a substantial part of finite-sample KS discrepancy, but observed component-duration, amplitude, overlap and count differences remain above it. This supports a restricted conclusion of incomplete component fidelity. An $n=120$ resampling reference is supplied for planning; it is not an evaluation of 120 generated minimum-jerk fits and is not substituted for the $n=10$ comparison.

7 Limitations and conclusions

The strategy-inclusive CVAE is a useful experimental model, with lower distribution distances than spline at matched n and stronger capacity at $n=8$. Spline remains the stronger deterministic shape compressor. A common-head comparison retains a smaller initiation-time advantage, but does not establish superiority on movement timing or isolate the latent representation from its training objective. Broad prediction of participant-level distribution parameters, a uniquely beneficial task condition, and stable interpretable strategy coordinates are not established.

The 357 zero-initiation labels (7.5%) occur because the post-go threshold is already satisfied; none is a missing-onset fallback. Among these, 109 have at least 50 ms of continuous pre-go threshold exceedance in filtered planar speed, and 100 do in raw planar speed. These observations cannot be dismissed as noise, but do not establish anticipation because event alignment and zero-phase filtering require review. The 74.6% share sometimes associated with these trials is a share of centered log-target variance, not measured trained-model loss or proof of the cause of timing extrapolation. Figure A2 discloses the original timing tails,

including CVAE initiation predictions reaching about 11.5 s; no post-hoc clipping is used to improve the benchmark.

No learning-curve experiment in this report establishes that sample size caused any model failure. More participants may help, but this remains a hypothesis. The four-fold evaluation measures performance within the available cohort, with repeated fitting rather than a fully independent replication. The report's distance reevaluation, event audit, behavioural-probe aggregation, common timing heads and sampling-reference analysis are post-review analyses. They reuse frozen main checkpoints and do not alter the canonical labels or exclusions. The source training records include dirty-worktree flags for two model families; checkpoint hashes and per-run source provenance are retained, rather than claiming every fit arose from a clean commit.

7.1 Points requiring domain confirmation

1. Confirm that the target-motion event, rather than appearance, is the permitted movement cue, and confirm its alignment with tracker frames and MAT pressedTime. Should the observed pre-go motion change trial validity or onset measurement?
2. Confirm whether the condition-2 fixation warning should affect inclusion, and whether the 1 s late-arrival cutoff is appropriate. These trials have not been silently removed.
3. Confirm tracker units, the x-y plane, onset threshold and 10 Hz filtering. Advise on minimum-jerk duration bounds, candidate orders and order criterion before any cognitive interpretation.

The immediate next step is to resolve these acquisition and analysis assumptions, then rerun affected stages. A different latent architecture or clinical interpretation is not required for the current workshop objective. We welcome comments on the modeling question, evaluation and scope of the conclusions.

Acknowledgments and reproducibility

We thank Moni Shahar for guidance on the project design and Prof. Jason Friedman for the experiment, data and methodological advice. AI assistance was used in software development and report preparation. Responsibility for verification and interpretation remains with the authors. Original outputs are preserved separately from the versioned corrected evaluation; Appendix B identifies the current artifacts.

References

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- [5] Flash T, Hogan N. The coordination of arm movements: an experimentally confirmed mathematical model. *J Neurosci.* 1985;5:1688-1703. doi:10.1523/JNEUROSCI.05-07-01688.1985.
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Appendix A: Representative outputs and diagnostic figures

Figure A1. Held-out reconstruction and context-fingerprint generation (fold 0, seed 42)

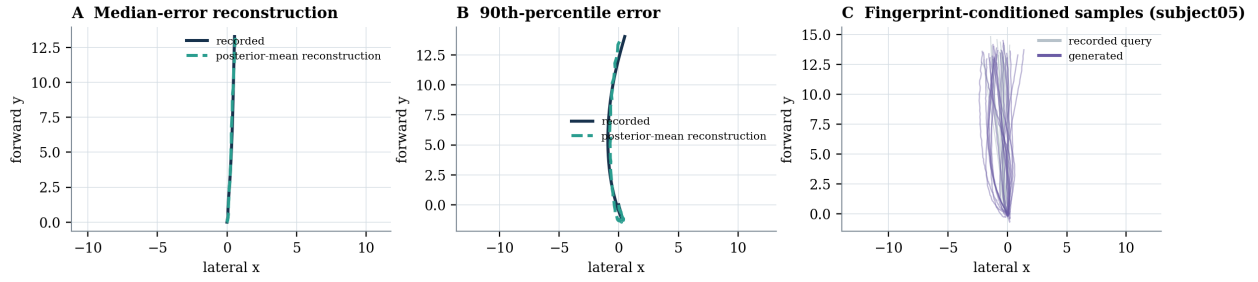


Figure A1. Held-out posterior-mean reconstructions and context-fingerprint generation. CVAE $n=8$, fold 0, seed 42; typical trial subject13_li_2_1_2_7; high-error trial subject19_li_2_1_1_8. Examples were chosen using reconstruction-error quantiles and do not replace all-participant evaluation.

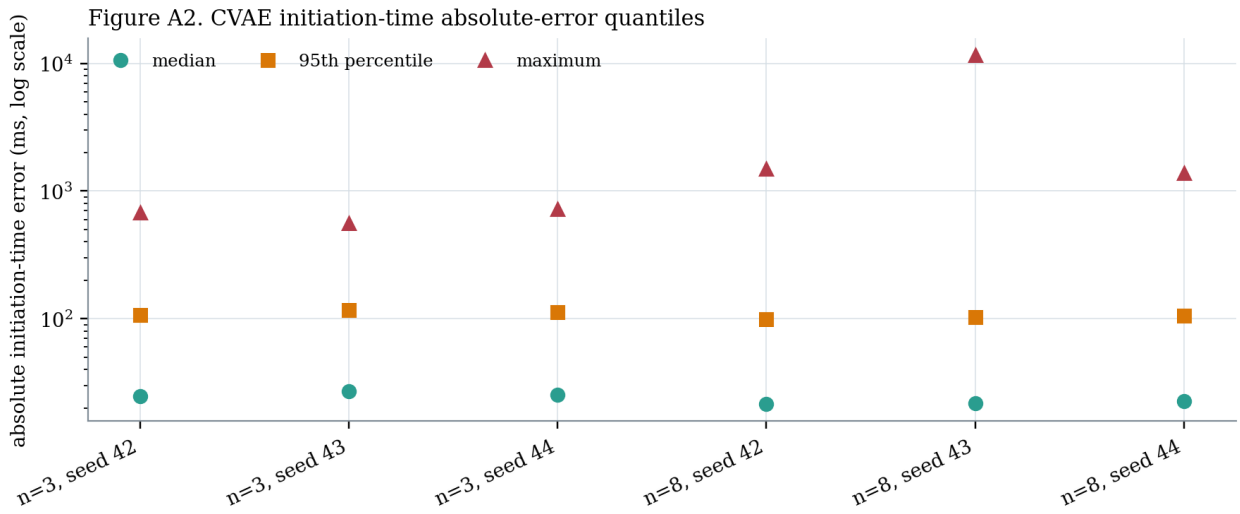


Figure A2. Original CVAE timing-error quantiles and maxima. Small median absolute errors can coexist with negative R-squared when a few errors are very large.

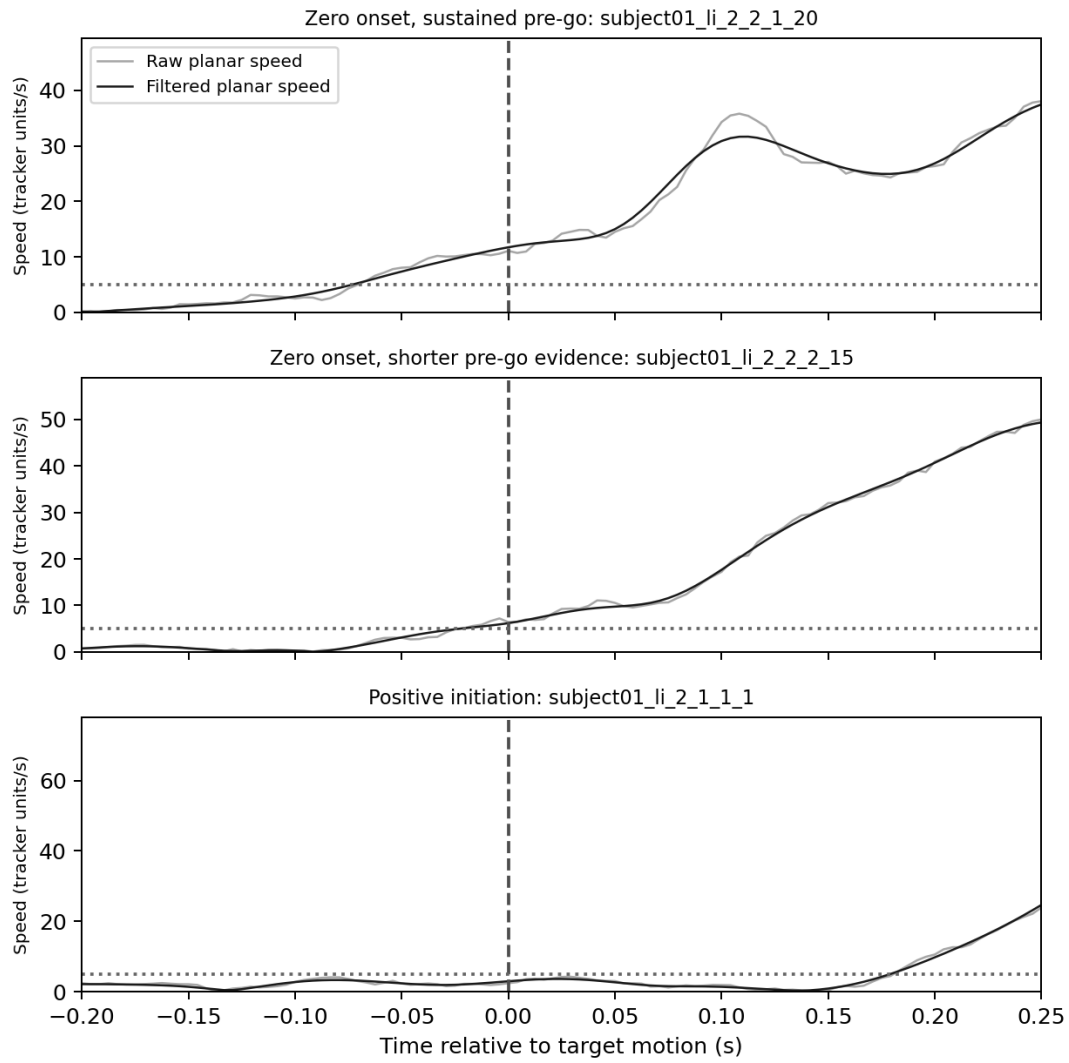


Figure A3. Event-audit examples with raw and filtered motion around the go event. Selection is deterministic within the audit groups; threshold exceedance is not a diagnosis of premature movement.

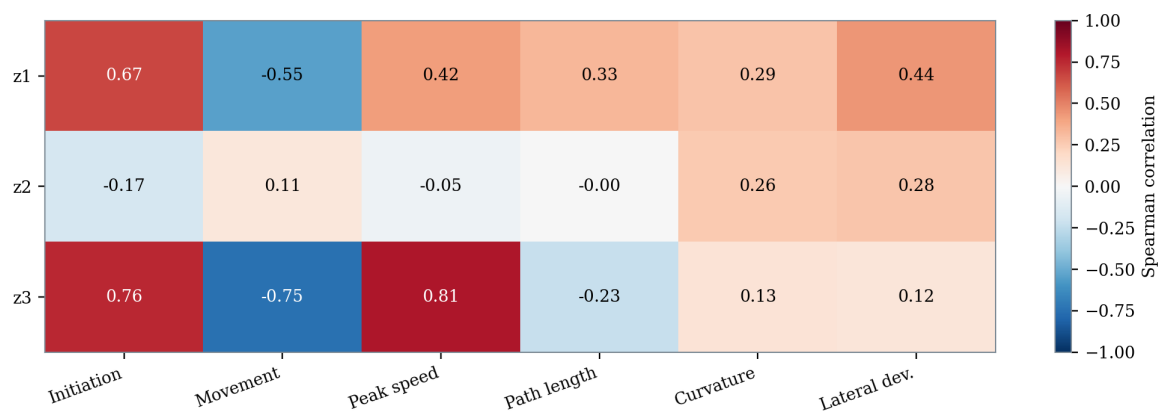


Figure A4. Decoder association heatmap, CVAE $n=3$, fold 0, seed 42. 500 draws from the training-centred shared latent covariance; fixed $sp=2$, $side=1$ and target speed 0.635 screen units/s. Spearman correlations are within this generated sample, not independent tests of causal control or human strategy.

Appendix B: Complete comparison and accompanying artifacts

Table B1. Complete dimension sweep. See Table 2 for metric definitions.

Model	n	MSE	Init. MAE (ms)	Move MAE (ms)	Mean KS	MMD ²	Enroll. (%)
Condition Ridge	-	1.234	98.5	72.7	0.277	0.126	-
Conditional AE	3	0.170	37.8	44.1	0.288	0.141	39.0
Conditional AE	8	0.026	39.9	42.7	0.259	0.111	46.5
CVAE	2	0.325	39.8	46.8	0.310	0.163	33.0
CVAE	3	0.177	37.8	43.4	0.278	0.131	40.9
CVAE	4	0.102	35.7	44.8	0.263	0.113	44.3
CVAE	8	0.032	34.8	44.2	0.219	0.067	50.7
Spline+PCA	2	0.197	56.6	50.9	0.355	0.214	29.6
Spline+PCA	3	0.097	56.2	51.2	0.325	0.181	36.4
Spline+PCA	4	0.076	56.2	51.2	0.302	0.167	38.4
Spline+PCA	8	0.023	55.6	50.2	0.255	0.115	48.6
Unconditional VAE	3	0.148	37.9	42.9	0.285	0.130	40.0
Unconditional VAE	8	0.033	36.0	42.2	0.215	0.064	50.8

The ZIP contains the scientific report, a separate student results guide, one Streamlit dashboard and machine-readable tables. The dashboard uses strategy-window CVAE checkpoints at n=2,3,4,8 (fold 0, seed 42); benchmark tables use all four folds and all saved seeds. Live generation is an illustrative checkpoint, not an ensemble. Minimum-jerk validation is available only for n=3 and n=8.

Extract the ZIP, install requirements_dashboard.txt in a Python environment, then run `python -m streamlit run src/confirmatory_dashboard.py` from the extracted folder. In Generate, select n and a context-enrolled participant, then adjust latent values and task condition. Held-out validation shows empirical/query comparisons; Benchmarks separates reconstruction and distribution metrics; Latent associations shows the current decoder's correlations; Diagnostics exposes timing, conditioning and secondary fitting limits. Condition controls are exploratory, not validated causal interventions.

Current corrected tables are under studies/review_corrected_evaluation/results: analysis (full 56 paired comparisons, feature-wise metrics, seed variation and timing tails); behavioral_probes (all available mean and mean-plus-SD scores and predictions); timing_fairness (validation factors, common-head predictions and paired tests); event_audit (all 4,732 trial rows); and sampling_reference (matched-size reference distributions). Unchanged training checkpoints and minimum-jerk fits remain under studies/final_strategy_evaluation. Small review tables and the dashboard assets accompany the compact ZIP; raw recordings and the full training matrix are not included.